

Titanic Dataset - Exploratory Data Analysis Report

Author: Mahnoor | Track: Zynxis Internship - Data Analysis

1. Executive Overview & Data Integrity

This report summarizes the Exploratory Data Analysis (EDA) conducted on the combined Titanic dataset (892 total records across train and test sets). Data integrity audits confirmed 0 missing values across all 16 features. A single duplicate row was identified and removed during the cleaning pipeline.

2. Key Analytical Findings

- Gender Protocol ('Women and Children First'):

Gender proved to be the strongest overall survival predictor ($r = -0.54$). Female passengers achieved a ~74% survival rate compared to only ~19% for male passengers.

- Socio-Economic Ticket Class Privilege:

First-class passengers (Pclass_1) experienced a ~63% survival probability. In contrast, third-class passengers (Pclass_3) saw a survival rate of just ~24% due to lower deck positions.

- Fare & Location Correlations:

Higher ticket fares correlated positively with survival outcomes ($r = +0.26$), reflecting proximity to upper-deck lifeboats and evacuation prioritization.

- Family Size Dynamics:

Moderate family sizes benefited from shared assistance during evacuation, whereas solo travelers and very large family units faced lower overall survival rates.

3. Conclusion

Demographic factors (Gender and Class) were the dominant drivers of survival on the Titanic. The dataset is fully cleaned, transformed, and structured for downstream predictive modeling.